The Supplier Evidence Maturity Model (SEMM)

A practical framework for EU-buyer-readiness of Brazilian exporters under EUDR, CBAM and CSDDD

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Abstract

Brazilian exporters increasingly discover that *market access is not the same as compliance evidence*: a shipment can clear customs while the supplier still cannot produce the buyer-readable, defensible documentation that European due-diligence, carbon and deforestation rules push down the supply chain. This report introduces the **Supplier Evidence Maturity Model (SEMM)** — a five-level self-assessment framework (Level 0 to Level 4) crossed with seven evidence categories — that lets a Brazilian supplier, or a European buyer assessing that supplier, locate its current evidence position and identify the next concrete step toward buyer-readiness. The model is instrument-agnostic by design: rather than tracking the shifting text of each regulation, it organises the *operational evidence* those regulations tend to require in common (identity and traceability, origin and geolocation, chain of custody, environmental destination, data security, carbon accounting, and governance). The contribution is a reusable diagnostic artifact — a maturity matrix and an accompanying checklist — intended to reduce the recurring gap between *having a process* and *being able to prove it to a buyer*.

Keywords: supplier evidence; ESG supply chain due diligence; EUDR compliance; CBAM; CSDDD; supplier traceability; buyer-readiness; audit-grade documentation; Brazil; EU-Mercosur.

1. Introduction

A recurring pattern in EU–Brazil trade is that a Brazilian supplier believes it is "compliant" because its goods enter the European market, while the European buyer — who carries the regulatory and reputational exposure — cannot obtain from that supplier the **documentation it can actually read, trust and defend** to its own auditors, regulators and board. The problem is not usually the absence of good practice on the ground. It is that good practice, when it is not captured as **structured, buyer-readable evidence**, does not survive the trip up the supply chain.

This report builds on a series of prior technical records on the EU–Brazil supplier evidence gap (see *References*) and moves from *diagnosis* to *tooling*. Earlier records argued that market access is not compliance and that regulatory clarity is not, by itself, audit-grade evidence. Here the aim is narrower and more practical: to give a supplier or a buyer a **structured way to measure where a given supply relationship stands** and what the next step is.

The framework deliberately does **not** attempt to restate the current legal text of each instrument. Those texts move — timelines have been postponed and simplification packages have been proposed and debated. What is more stable is the **class of operational evidence** that European buyers ask their suppliers to produce, regardless of which specific instrument is driving the request in a given quarter. The SEMM is built on that more stable layer.

2. Scope and method

The model was developed by structured synthesis of (a) the operational documentation categories that recur across European supply-chain instruments; (b) the practical failure modes observed when a supplier holds a process but cannot evidence it; and (c) established maturity-model design (discrete, ordered levels; observable criteria at each level; no level skipped in practice). It is a **conceptual and organisational** contribution, not an empirical study; it presents no primary statistics and makes no claim about the frequency of any condition in the Brazilian exporter population.

Three European instruments are used as orientation because they are the ones most frequently cited to Brazilian exporters:

- **EUDR** — the EU deforestation-free products Regulation, Regulation (EU) 2023/1115. *Application dates by operator size must be verified on EUR-Lex; these were amended after adoption.*
- **CBAM** — the Carbon Border Adjustment Mechanism, Regulation (EU) 2023/956, with a transitional period followed by a definitive regime. *The current phase and reporting obligations must be verified.*
- **CSDDD** — the Corporate Sustainability Due Diligence Directive, Directive (EU) 2024/1760, transposed and phased in by member states. *Transposition and phase-in dates must be verified, as they have been subject to review (including simplification / "Omnibus" discussions).*

The **EU–Mercosur** agreement is noted as background context that may raise, not lower, the salience of supplier evidence; its status is political/legislative and *must be verified* before being relied upon. Nothing in this report should be read as advice on whether a specific instrument applies to a specific company. That determination is legal and case-specific.

3. The core distinction: process, compliance, and evidence

The model rests on separating three things that are routinely conflated:

1. **Process** — what the supplier actually does (e.g. it segregates material, it geolocates plots, it sanitises data-bearing devices).
2. **Compliance** — whether that process meets an applicable requirement.
3. **Evidence** — whether the supplier can *produce documentation that a buyer can read, trust and defend* showing 1 and 2.

A supplier can have (1) without (3). European buyers increasingly act on (3), because (3) is what they must hand to their own regulators and auditors. The SEMM measures (3), on the working assumption that (1) exists — it measures **provability**, not virtue.

Three quality attributes define buyer-usable evidence throughout this report:

- **Buyer-readable** — intelligible to the buyer without access to the supplier's internal systems or tacit knowledge.

- **Audit-grade** — traceable to a source, dated, attributable, and consistent over time (a concept, not an audit opinion).
- **Regulatory-defensible** — capable of supporting the buyer's own account to a competent authority.

4. The Supplier Evidence Maturity Model

The SEMM places a supply relationship on one of five levels. Levels are **ordered and cumulative**: a level is reached only when its criteria are met for the evidence categories in Section 5.

Level	Name	Defining condition
0	Opaque	No structured evidence. Claims rest on trust, informal assurance or verbal confirmation. Nothing survives a buyer's document request.
1	Documented	Evidence exists but is scattered, inconsistent, and produced reactively per request. Documents may be genuine yet not buyer-readable (internal codes, no dates, no attribution).
2	Structured	Evidence is organised by category, consistently formatted, dated and attributable. A buyer can obtain a coherent file on request within a defined time.
3	Buyer-ready	Evidence is proactively maintained, buyer-readable and regulatory-defensible, mapped to the categories a buyer asks about, and refreshable without a special project.
4	Continuously assured	Evidence is maintained as a living system, periodically self-checked, versioned, and resilient to changes in personnel or systems. Gaps are detected internally before a buyer finds them.

The practical target for most exporters is **Level 3**. Level 4 is appropriate where the buyer relationship is strategic or where the supplier serves multiple regulated buyers and wants to answer all of them from one system.

5. Evidence categories

The columns of the model are the **seven categories** of evidence that recur across European supply-chain requests. Not every category applies to every product; the supplier assesses only those relevant to what it sells.

1. **Identity & traceability** — the ability to tie a delivered lot back to its supplier, unit and, where relevant, its constituent inputs.
2. **Origin & geolocation** — where the input physically came from (for EUDR-type requests, plot- or site-level geolocation of production).
3. **Chain of custody** — unbroken, documented custody from origin to delivery, so that identity and origin are not lost at a hand-off.
4. **Environmental destination / end-of-life** — for materials, by-products or returned assets: documented, verifiable destination rather than an isolated certificate.
5. **Data security** — for data-bearing assets and IT equipment: documented sanitisation and media governance.
6. **Carbon / CO_2e accounting** — emissions data attributable to the goods, at the granularity a buyer (or a CBAM-type request) needs, with a stated method.
7. **Governance & accountability** — who is responsible, how the evidence is controlled, and how errors are corrected — the layer that makes the other six defensible.

6. The self-assessment matrix

The citable artifact of this report is the **matrix**: each *relevant category* (Section 5) is scored against the *five levels* (Section 4). The relationship's overall level is the **lowest** level reached across its relevant categories — because a buyer's exposure is set by the weakest evidenced link, not the average.

Evidence category	L0 Opaque	L1 Documented	L2 Structured	L3 Buyer-ready	L4 Assured
Identity & traceability	☐	☐	☐	☐	☐
Origin & geolocation	☐	☐	☐	☐	☐
Chain of custody	☐	☐	☐	☐	☐
Environmental destination	☐	☐	☐	☐	☐
Data security	☐	☐	☐	☐	☐
Carbon / CO_2e	☐	☐	☐	☐	☐
Governance & accountability	☐	☐	☐	☐	☐

How to use it. (1) Mark each relevant category "not applicable" or score it 0–4. (2) The overall position is the lowest score among applicable categories. (3) The next action is whatever moves that lowest category up one level — not a broad programme. This keeps remediation **sequenced and affordable**, which matters for small and mid-size exporters.

7. Application notes by driver

The matrix is instrument-agnostic, but the categories that dominate differ by what is driving the buyer's request:

- **Deforestation-driven (EUDR-type) requests** load heavily on *origin & geolocation* and *chain of custody* — the buyer must connect the delivered commodity to a non-deforested plot. *Verify current EUDR scope/dates before acting.*
- **Carbon-driven (CBAM-type) requests** load on *carbon / CO_2e accounting* with a stated, consistent method at the required granularity. *Verify the current CBAM phase.*
- **Due-diligence-driven (CSDDD-type) requests** load on *governance & accountability* and pull the other categories in as the evidence a due-diligence process must rest on. *Verify transposition / phase-in.*

In practice a Brazilian exporter serving several European buyers faces a **union** of these requests, which is the argument for a single evidence system (Level 3–4) rather than one improvised response per buyer.

8. Implementation roadmap

A pragmatic path from Level 1 to Level 3:

1. **Map relevant categories** for the actual product and buyers (drop the non-applicable ones honestly).
2. **Score today's position** with the matrix; identify the lowest applicable category.
3. **Fix the format before the substance** — much Level-1 evidence is genuine but not buyer-readable (undated, internal codes, no attribution). Making it readable is often the cheapest level jump.

4. **Sequence one category at a time**, lowest first, to reach an even Level 2 across all applicable categories.
5. **Move to Level 3** by making the evidence proactive and refreshable — a maintained file, not a project reopened per request.
6. **Assign governance** (category 7) so the system survives staff turnover.

9. Limitations

This is a conceptual framework, version 1.0, not peer reviewed and not empirically validated against a sample of exporters. It presents no primary statistics. Level definitions are qualitative and will benefit from field calibration. The regulatory references are orientation only and are subject to amendment; none of the flagged items should be relied upon without verification against current official sources. The model measures *provability of evidence*, not the underlying environmental or social performance, and is not a substitute for legal, customs, technical or assurance advice.

10. How to cite

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References

Regulatory instruments are cited for orientation; verify exact titles, numbers and dates on EUR-Lex.

1. Regulation (EU) 2023/1115 — deforestation-free products (EUDR). *Verify application dates.*
2. Regulation (EU) 2023/956 — Carbon Border Adjustment Mechanism (CBAM). *Verify current phase.*
3. Directive (EU) 2024/1760 — Corporate Sustainability Due Diligence (CSDDD). *Verify transposition / phase-in.*
4. EU–Mercosur Partnership Agreement — political/legislative background. *Verify status.*
5. Villanova, M. (2026). Prior technical records on the EU–Brazil supplier evidence gap, buyer-readiness and audit-grade documentation, deposited on Zenodo. Full list with DOIs: villanovaesg.com/publications.